Existence of LU and LDL^T Factorizations without Pivoting

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Abstract

We give a unified treatment of two existence questions for triangular factorizations without pivoting over $\mathbb{R}$: when does a square matrix A admit a factorization $A = LU$ with L lower triangular and U upper triangular, and, when A is symmetric, a factorization $A = LDL^T$ with L lower triangular and D diagonal? No permutations are allowed, A may be singular, and zeros are allowed on the diagonals of all factors; this distinguishes the problem from the classical theory, which assumes that all leading principal minors are nonzero. Both questions are answered by nullity inequalities comparing each leading principal submatrix with the corresponding leading block of columns and rows. The unsymmetric criterion recovers the theorem of Okunev and Johnson; the symmetric criterion appears to be new. Both sufficiency proofs follow a single strategy: reduce A by an invertible triangular equivalence (a congruence in the symmetric case) to a matching normal form, observe that the criterion is invariant under such reductions and becomes a combinatorial prefix condition on the matching, and then factor the normal form explicitly by a sorted slot assignment, a prefix form of Hall's theorem: each matched entry — and, in the symmetric case, each half of an off-diagonal pair split into a difference of two symmetric rank-one terms — is placed at an earlier diagonal slot, with isolated zero indices supplying the spare slots.

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1. Introduction

Let $A \in \mathbb{R}^{n\times n}$. It is classical that when A is nonsingular, a factorization $A = LU$ with L lower triangular and U upper triangular exists if and only if every leading principal submatrix $A[1{:}k, 1{:}k]$ is nonsingular [1, 2]. The standard remedy when this condition fails is pivoting: for any A there are permutations P (and Q) such that $PA = LU$ or $PAQ = LU$. Pivoting, however, changes the problem. When the row and column indices carry meaning — temporal order in autoregressive models, causal order in structural equation models, or a fixed elimination order imposed by an application — the factorization must respect the original indexing, and the correct question becomes: *for which matrices A does $A = LU$ hold with no permutations at all?*

Two features make the unpivoted question fundamentally different from the classical one. First, A may be singular, or may have singular leading principal blocks, and yet admit a perfectly good factorization. Second, the factors must then be allowed to carry *zeros on their diagonals*: we require only that L be lower triangular and U upper triangular, not that they be invertible. The classical minor criterion says nothing in this regime. For example,

$$A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \tag{1.1}$$

are both symmetric with singular leading blocks. The swap matrix A_1 has no factorization $A_1 = LU$ (from $a_{11} = l_{11}u_{11} = 0$, either the first row or the first column of LU must vanish). Yet the larger, more degenerate matrix A_2 factors:

$$A_2 = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}}_{LU} = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}}_{LDL^T}. \tag{1.2}$$

Both factorizations require zero diagonal entries: the leading zero index of A_2 acts as a resource that later off-diagonal structure can be routed through. Understanding exactly when such routing is possible is the subject of this paper. Note also that D displays the rank and the inertia of A_2: it has two nonzero entries ($\operatorname{rank} A_2 = 2$), one positive and one negative (A_2 has eigenvalues $1, -1, 0$). This is a property of a well-chosen factorization, not of all of them — when L is singular, Sylvester's law of inertia does not constrain D (Example 9.1) — and Theorem 8.2 shows that a rank- and inertia-revealing choice always exists.

We answer the question for both the unsymmetric and symmetric forms of the problem. Write $\mathrm{null}(X) = (\#\mathrm{cols~of~} X) - \mathrm{rank}(X)$ for the right nullity. Our two main results are the following (Theorems 4.1 and 4.2):

- $A \in \mathbb{R}^{n\times n}$ admits a factorization $A = LU$ with L lower triangular and U upper triangular, zeros allowed on both diagonals, if and only if for every $k = 1, \ldots, n$,

$$\mathrm{null}(A[1{:}k, 1{:}k]) \leq \mathrm{null}(A[:, 1{:}k]) + \mathrm{null}(A[1{:}k, :]^T).$$

- A symmetric $A \in \mathbb{R}^{n\times n}$ admits a factorization $A = LDL^T$ with L lower triangular, zeros allowed on the diagonal, and D diagonal if and only if for every $k = 1, \ldots, n$,

$$\mathrm{null}(A[1{:}k, 1{:}k]) \leq 2\ \mathrm{null}(A[:, 1{:}k]).$$

The first criterion is equivalent to the rank inequality of Okunev and Johnson [3, 4]. The second criterion, and the fact that for symmetric matrices the two criteria coincide — so that a symmetric matrix admits an unpivoted LU factorization if and only if it admits an unpivoted LDL^T factorization — appear to be new.

The unified method. Both sufficiency proofs follow the same three-step strategy.

1. *Reduce.* Every A can be written $A = M\Pi N$ with M invertible lower triangular, N invertible upper triangular, and Π a matching matrix. Every symmetric A can be written $A = MPM^T$ with M invertible lower triangular and P a signed matching matrix whose pair blocks have the normalized real form $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and whose diagonal entries lie in $\{0, \pm 1\}$ (Lemmas 5.3 and 5.4).
2. *Transfer.* The nullities appearing in the criteria are invariant under these triangular equivalences and congruences, so the criteria may be checked on the normal forms. On normal forms they become prefix conditions on matchings (Lemmas 6.2 and 6.4).
3. *Assign.* A matching matrix satisfying the prefix condition is factored by assigning each matched entry a diagonal slot $t \leq \min(i, j)$. In the symmetric case a matched entry cannot be realized alone: each off-diagonal pair splits into a difference of two symmetric rank-one terms and consumes *two* slots, and the prefix condition supplies exactly enough earlier slots, the spares being created by isolated zero indices (Theorems 7.1 and 7.3).

Necessity is a short rank argument, and the standard normal-form reductions and invariance facts are relegated to the appendices so that the main text can focus on the new prefix-condition and slot-assignment mechanisms.

Contributions.

- A unified normal-form proof of unpivoted LU existence and unpivoted LDL^T existence for real matrices with possibly singular triangular factors.
- A necessary and sufficient condition for the existence of an unpivoted LDL^T factorization with a truly diagonal D and possibly singular L, together with the corollary that for symmetric matrices unpivoted LU and LDL^T existence are equivalent.
- An explicit combinatorial mechanism for the symmetric problem: the prefix condition supplies, for every off-diagonal pair, the two earlier slots required by its rank-one splitting, with isolated zero indices acting as the spare slots.
- Canonical sparsity and rank-revealing structure (Section 8): both constructions can be steered to factorizations with exactly $\operatorname{rank}(A)$ populated columns, whose sparsity envelope is determined by the rank-jump matching of A and is simultaneously optimal, over all unpivoted factorizations, for every leading block at once (Theorems 8.1 and 8.2 and proposition 8.3).

Organization. Section 2 positions the paper. Section 3 fixes notation, and Section 4 states the main theorems. Section 5 states the matching normal forms. Section 6 converts the criteria into prefix conditions, Section 7 factors the normal forms, Section 8 develops the sparsity and rank-revealing structure of the factors, and Section 9 assembles the proofs. The appendices contain the standard rank facts, necessity and transfer arguments, normal-form proofs, and an additional example.

2. Related work

Unpivoted LU for nonsingular matrices. The classical existence theory assumes invertibility. If A is nonsingular, then $A = LU$ exists if and only if all leading principal submatrices are nonsingular, in which case L and U are invertible and the factorization with unit lower triangular L is unique; see Higham [1] for a survey and [2] for textbook treatments. Higham notes explicitly that when a proper leading principal submatrix is singular a factorization "may exist, but if so it is not unique," without characterizing when. Lau and Markham [5] and Johnson, Olesky, and van den Driessche [6] showed that a possibly singular A has a *unique* unit LU factorization precisely when the $n-1$ *proper* leading principal minors are nonzero; the analysis of [6] (inheritance of entries, fill-in) assumes this condition throughout and does not address existence when it fails.

Unpivoted LU for arbitrary matrices. The existence question for arbitrary, possibly singular A was settled by Okunev and Johnson. Their result, circulated as a 1997 manuscript and posted as [3], recently appeared as [4]: over a field, A admits $A = LU$, with no invertibility requirement on the factors, if and only if

$$\operatorname{rank}(A[1{:}k, 1{:}k]) + k \;\geq\; \operatorname{rank}(A[1{:}k, :]) + \operatorname{rank}(A[:, 1{:}k]), \qquad k = 1, \ldots, n. \tag{2.1}$$

Their sufficiency proof is an induction on "factor complements" (generalized Schur complements), choosing the first column of L and the first row of U case by case; the published version also develops "near-LU" factorizations with almost-triangular factors when (2.1) fails, and re-derives PLU and three-factor decompositions. Darve [7] restates (2.1) in the nullity form used in this paper, gives a different constructive proof based on *restricted pivoting* (only rows or columns that are zero in the active Schur complement are moved, so that undoing the internal permutations preserves triangularity), and develops rank-revealing structure, tight sparsity bounds for the factors, and criteria for one-sided unit-triangular factors. The present paper gives a third, normal-form proof of the same criterion, whose main advantage is that it extends verbatim to the symmetric LDL^T problem.

Dopico, Johnson, and Molera [8] study the related question in which L is required to be unit lower triangular (hence invertible): existence is characterized through a canonical form of A under left multiplication by unit lower triangular matrices (the rank-revealing minimal canonical form), and, importantly, all LU factorizations of a singular matrix are parametrized, quantifying their non-uniqueness. Our normal form differs in that the reduction acts on both sides (or by congruence), and the middle factor — not an upper triangular matrix — carries the combinatorial obstruction. Nagarajan, Devasahayam, and Soundararajan [9] showed that *every* square matrix over a field is a product of three triangular matrices LUM; the two-factor existence question is exactly the question of when the third factor can be dispensed with. Jeffrey [10] standardizes full-rank LU factorings of rectangular and rank-deficient matrices in the form $P_r L D^{-1} U P_c$; since row and column permutations are built in, existence is unconditional and the unpivoted question does not arise. For the structured class of M-matrices, McDonald and Schneider [11] gave graph-theoretic necessary and sufficient conditions for a singular M-matrix to admit an unpivoted (block) LU factorization into possibly singular M-matrix factors. Rank-revealing LU with pivoting chosen for numerical size, rather than algebraic existence, is developed by Miranian and Gu [12], and the conditioning of the triangular factors of pivoted factorizations is analyzed by Stewart [13]. General surveys of matrix factorizations include [14].

Symmetric factorizations. For symmetric positive semidefinite matrices over $\mathbb{R}$, an unpivoted Cholesky-type factorization always exists even in the singular case: $A = LL^T$ with

L lower triangular, possibly with zero diagonal entries; see Higham [15] for a detailed treatment. Our symmetric criterion is consistent with this: positive semidefiniteness forces $\text{null}(A[1{:}k, 1{:}k]) = \text{null}(A[:, 1{:}k])$ for all k, so the criterion of Theorem 4.2 holds automatically (Corollary 4.5). For symmetric *indefinite* matrices the classical approach abandons the diagonal D: with symmetric pivoting, $PAP^T = LDL^T$ always exists when D is allowed to be block diagonal with 1×1 and 2×2 blocks, as introduced by Bunch and Parlett [16] and made efficient by Bunch and Kaufman [17]; a rank-profile-revealing variant over arbitrary fields, still with symmetric pivoting and with 2×2 antidiagonal blocks in D, is given by Dumas and Pernet [18]. Closest in spirit to our symmetric reduction is the elegant paper of Van Dooren and Strang [19], which factors a real symmetric matrix as $A = LPL^T$ *without pivoting*, where P is a signed-matching middle factor built from diagonal entries $0, \pm 1$ and symmetric exchange blocks; our Lemma 5.4 can be read as a real normalized version of their reduction. Neither [19] nor [18] addresses when the middle factor can be made *truly diagonal*, which is exactly the step from $A = MPM^T$ to $A = LDL^T$ taken in Sections 6 and 7. The 2×2 blocks of Bunch–Parlett reappear in our theory in a different role: they are the irreducible blocks of the signed matching normal form, i.e., the obstruction that must be routed through earlier zero indices rather than a device enabled by pivoting. We are not aware of a previously published necessary and sufficient condition for the unpivoted diagonal-D problem with possibly singular L.

Triangular canonical forms and the Bruhat decomposition. Our reduction lemma states that every A equals $M\Pi N$ with M, N invertible triangular and Π a matching matrix, and that the matching is an invariant of A. For invertible A this is a form of the Bruhat decomposition $A = LPU$, whose permutation P is uniquely determined by A; see Elsner [20] and Strang [21, 22] for elimination-based accounts, and [8] for the related one-sided canonical form. For singular matrices the permutation becomes a partial permutation: this is the generalized Bruhat (or LEU) decomposition [23], and the invariant matching is the *rank profile matrix* of Dumas, Pernet, and Sultan [24], with which part (1) of our Proposition 5.5 coincides. On the symmetric side, Szechtman [25] classified symmetric matrices under congruence by triangular groups, and Bagno and Cherniavsky [26] studied the corresponding rank-control invariants of congruence B-orbits over $\mathbb{C}$. Our Lemma 5.4 and proposition 5.5 thus recover known canonical forms; we include short elimination-style constructive proofs to keep the paper self-contained and to fix the normalization used in the symmetric slot assignment argument.

Positioning. In summary: the unsymmetric criterion in Theorem 4.1 is due to Okunev and Johnson [3, 4], with alternative proofs and structural refinements in [7]; the triangular canonical forms we use connect to the generalized Bruhat decomposition [20, 23, 24] and, in the symmetric case, to Van Dooren–Strang [19] and Szechtman [25]. The contributions of the present paper are the unified normal-form route to *existence* theorems, the symmetric LDL^T criterion (Theorem 4.2) together with the equivalence of LU and LDL^T existence for symmetric matrices (Corollary 4.3), the prefix-condition combinatorics and slot-assignment constructions of Sections 6 and 7.

3. Notation

Throughout, matrices have real entries. We use the slicing notation standard in numerical linear algebra [2]: $A[i{:}j, k{:}l]$ is the submatrix of A with rows i through j and columns k through l; a bare colon denotes the full range, so $A[:, 1{:}k]$ is the block of the first k columns and $A[1{:}k, :]$ the block of the first k rows. We write e_i for the ith standard basis vector, I_n for the identity, and u^T for the transpose.

For an $m \times p$ matrix X, $\mathrm{rank}(X)$ is its rank and

$$\mathrm{null}(X) = p - \mathrm{rank}(X)$$

is the dimension of its *right* null space. In particular, for the three blocks entering our criteria — each of which has exactly k columns — we have

$$\mathrm{null}(A[1{:}k, 1{:}k]) = k - \mathrm{rank}(A[1{:}k, 1{:}k]), \qquad \mathrm{null}(A[:, 1{:}k]) = k - \mathrm{rank}(A[:, 1{:}k]),$$
$$\mathrm{null}(A[1{:}k, :]^T) = k - \mathrm{rank}(A[1{:}k, :]).$$

A matrix L is *lower triangular* if $L[i, j] = 0$ for $i < j$, and U is *upper triangular* if $U[i, j] = 0$ for $i > j$. We emphasize that triangular factors in this paper are *not* assumed invertible: zeros are allowed on their diagonals. A triangular matrix is invertible if and only if its diagonal entries are all nonzero, in which case its inverse is triangular of the same kind, and products of lower (resp. upper) triangular matrices are lower (resp. upper) triangular. We will also use repeatedly that if M is lower triangular and N is upper triangular, then

$$M[1{:}k, :] = \begin{pmatrix} M_k & 0 \end{pmatrix}, \qquad N[:, 1{:}k] = \begin{pmatrix} N_k \\ 0 \end{pmatrix}, \tag{3.1}$$

where $M_k = M[1{:}k, 1{:}k]$ and $N_k = N[1{:}k, 1{:}k]$; if M (resp. N) is invertible then so is M_k (resp. N_k), since the leading block of a triangular matrix carries its first k diagonal entries.

The elementary rank facts used below, together with the short necessity and transfer arguments, are collected in Section Appendix A.

4. Main results

Theorem 4.1 (unpivoted LU). *Let $A \in \mathbb{R}^{n\times n}$. The following are equivalent:*

1. *there exist a lower triangular $L \in \mathbb{R}^{n\times n}$ and an upper triangular $U \in \mathbb{R}^{n\times n}$, with zeros allowed on both diagonals, such that $A = LU$;*
2. *for every $k = 1, \ldots, n$,*

$$\operatorname{null}(A[1{:}k, 1{:}k]) \;\le\; \operatorname{null}(A[:, 1{:}k]) + \operatorname{null}(A[1{:}k, :]^T). \tag{4.1}$$

By the rank–nullity relations of Section 3, (4.1) is equivalent to the rank form $\operatorname{rank}(A[1{:}k, 1{:}k]) + k \ge \operatorname{rank}(A[1{:}k, :]) + \operatorname{rank}(A[:, 1{:}k])$ of Okunev and Johnson [3, 4].

Theorem 4.2 (unpivoted LDL^T). *Let $A \in \mathbb{R}^{n\times n}$ with $A = A^T$. The following are equivalent:*

1. *there exist a lower triangular $L \in \mathbb{R}^{n\times n}$, with zeros allowed on the diagonal, and a diagonal $D \in \mathbb{R}^{n\times n}$ such that $A = LDL^T$;*
2. *for every $k = 1, \ldots, n$,*

$$\operatorname{null}(A[1{:}k, 1{:}k]) \;\le\; 2 \operatorname{null}(A[:, 1{:}k]). \tag{4.2}$$

Equivalently, in rank form, $\operatorname{rank}(A[1{:}k, 1{:}k]) + k \ge 2 \operatorname{rank}(A[:, 1{:}k])$ for all k. Note that (4.2) is precisely (4.1) specialized to a symmetric matrix: when $A = A^T$ we have $A[1{:}k, :]^T = A[:, 1{:}k]$, so the two terms on the right-hand side of (4.1) coincide. This gives immediately:

Corollary 4.3 (symmetric matrices: LU and LDL^T are equivalent). *A symmetric matrix $A \in \mathbb{R}^{n\times n}$ admits an unpivoted LU factorization if and only if it admits an unpivoted LDL^T factorization.*

Proof. If $A = LDL^T$ then $A = L\,(DL^T)$ is an LU factorization. Conversely, if $A = LU$ exists then (4.1) holds by Theorem 4.1, which for symmetric A is (4.2), and Theorem 4.2 produces an LDL^T factorization. $\square$

The next corollaries show that the criteria specialize to the familiar statements in the two regimes covered by the classical theory.

Corollary 4.4 (nonsingular case). *Let A be nonsingular. Then (4.1) holds if and only if all leading principal submatrices $A[1{:}k, 1{:}k]$ are nonsingular. In particular, Theorems 4.1 and 4.2 recover the classical criteria for nonsingular (symmetric) matrices.*

Proof. If A is nonsingular, its first k columns are linearly independent and its first k rows are linearly independent, so $\operatorname{null}(A[:,1{:}k]) = \operatorname{null}(A[1{:}k,:]^T) = 0$. Thus (4.1) reads $\operatorname{null}(A[1{:}k,1{:}k]) \le 0$, i.e., $A[1{:}k,1{:}k]$ is nonsingular, for every k. The same computation applies to (4.2). $\square$

Corollary 4.5 (positive semidefinite case). *Let A be symmetric positive semidefinite. Then (4.2) holds, so $A = LDL^T$ exists without pivoting; moreover* $\operatorname{null}(A[1{:}k,1{:}k]) = \operatorname{null}(A[:,1{:}k])$ *for every k.*

Proof. Write $A = B^TB$. Then $A[:,1{:}k] = B^T B[:,1{:}k]$ and $A[1{:}k,1{:}k] = B[:,1{:}k]^T B[:,1{:}k]$. For real B, $B[:,1{:}k]x = 0 \iff B[:,1{:}k]^T B[:,1{:}k]x = 0$ (multiply by x^T), so $\operatorname{rank}(A[1{:}k,1{:}k]) = \operatorname{rank}(B[:,1{:}k]) \ge \operatorname{rank}(A[:,1{:}k]) \ge \operatorname{rank}(A[1{:}k,1{:}k])$, where the last inequality holds because $A[1{:}k,1{:}k]$ consists of rows of $A[:,1{:}k]$. Hence $\operatorname{null}(A[1{:}k,1{:}k]) = \operatorname{null}(A[:,1{:}k]) \le 2\operatorname{null}(A[:,1{:}k])$. $\square$

Corollary 4.5 is consistent with the classical fact that a singular positive semidefinite matrix admits an unpivoted Cholesky factorization $A = LL^T$ with possibly zero diagonal entries in L [15]. Indefinite symmetric matrices, by contrast, may fail (4.2) — the swap matrix A_1 of (1.1) fails it at $k = 1$ — which is why the classical unpivoted theory for indefinite matrices resorts to block-diagonal D with symmetric pivoting [16, 17].

Remark 4.6 (zeros on the diagonals are essential). *If L is required to be invertible (e.g., unit lower triangular), the existence conditions become strictly stronger and different in kind; see [8] and [7, Sec. 3]. The matrix A_2 of (1.1) satisfies* (4.1) *and* (4.2) *but admits no factorization with L or U invertible. Indeed, if $A_2 = LU$ with L invertible, then $0 = A_2[1{:}2,1{:}2] = L[1{:}2,1{:}2]\,U[1{:}2,1{:}2]$ forces $U[1{:}2,1{:}2] = 0$, hence $U[:,1{:}2] = 0$ by (3.1) and $A_2[:,1{:}2] = L\,U[:,1{:}2] = 0$, contradicting $A_2[3,2] = 1$; the case of invertible U follows by transposing the symmetric A_2. Every factorization must instead route the exchange pair* $(2,3)$ *through the zero index* 1*, forcing zero diagonal entries, as in (1.2).*

Remark 4.7 (singular A, singular blocks). *Neither theorem places any restriction on* $\operatorname{rank}(A)$. *The nonsingular matrix $\left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)$ fails (4.1) at $k = 1$ (its leading entry vanishes while its first row and column are nonzero), whereas the rank-two matrix A_2 of (1.1) satisfies both criteria. Existence is governed not by how singular A is, but by whether every deficiency of a leading block is matched by global deficiencies of the corresponding rows and columns.*

5. Matching normal forms

The engine of the sufficiency proof is a reduction of A to a sparse normal form by invertible triangular transformations. In the unsymmetric case the normal form is a matching between row and column indices. In the symmetric real case, we use a triangular congruence to normalize every two-index block to one fixed representative.

Definition 5.1 (matching matrix). *A* matching matrix $\Pi \in \mathbb{R}^{n\times n}$ *is a partial permutation matrix: a* 0–1 *matrix with at most one* 1 *in each row and in each column. Equivalently,* $\Pi = D\Pi_0$ *for a permutation matrix* Π_0 *and a* 0–1 *diagonal matrix* D *— a permutation matrix with some rows zeroed out (or* $\Pi = \Pi_0 D'$*, with some columns zeroed out). We write*

$$\mathcal{M} = \mathcal{M}(\Pi) := \{(i,j){:}\Pi[i,j] = 1\}, \qquad \textit{so that} \qquad \Pi = \sum_{(i,j)\in\mathcal{M}} e_i e_j^T,$$

and call the elements of $\mathcal{M}$ matched pairs*; the defining property says precisely that no two matched pairs share a row index* i*, and no two share a column index* j*.*

Note that diagonal pairs (t,t) — the nonzero diagonal entries of Π — are allowed; they will correspond to the classical nonzero pivots.

Definition 5.2 (signed matching matrix). *A* signed matching matrix *is a symmetric matrix* $P \in \mathbb{R}^{n\times n}$ *with at most one nonzero entry in each row (equivalently, in each column), all of whose off-diagonal nonzero entries equal* 1 *and all of whose diagonal nonzero entries equal* ± 1.

Despite the parallel name, a signed matching matrix is *not* a special case of Definition 5.1: its diagonal entries may equal -1. This discrepancy is forced by the congruence action. A congruence rescales a diagonal entry γ_t by a nonzero *square* ($\gamma_t \mapsto s^2\gamma_t$ when index t is scaled by s), so $\operatorname{sgn}\gamma_t$ is a congruence invariant — it carries the inertia of A, as the example $A = (-1)$ already shows — and negative diagonal entries are therefore unavoidable in any normal form for congruence. The diagonal signs are also the *only* discrepancy: a signed matching matrix with no -1 entries is precisely a matching matrix that is symmetric, and $|\gamma_t|$ can be, and in Definition 5.2 has been, normalized to 1 (the one step in the paper that uses real square roots; everything else is rational in the entries of A).

Symmetry sorts the indices into three classes, for which we fix terminology: an index t with $P[t,t] = \gamma_t \in \{\pm 1\}$ is a *singleton*; indices $i < j$ with $P[i,j] = P[j,i] = 1$ form a *pair*, and the sparsity forces $P[i,i] = P[j,j] = 0$, so a pair carries the exchange block $P[\{i,j\},\{i,j\}] = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$; every remaining index is an *isolated zero* index, whose row and column vanish.

Lemma 5.3 (triangular equivalence to a matching matrix). *For every $A \in \mathbb{R}^{n\times n}$ there exist an invertible lower triangular M, an invertible upper triangular N, and a matching matrix Π such that*

$$A = M\,\Pi\,N.$$

The decomposition can be computed in $O(n^3)$ arithmetic operations.

When A is nonsingular, the factorization in Lemma 5.3 is the usual Bruhat decomposition. In the singular case, the permutation matrix is replaced by a partial permutation matrix, giving the generalized Bruhat, or rank profile, decomposition.

Lemma 5.4 (triangular congruence to a signed matching matrix). *For every symmetric $A \in \mathbb{R}^{n\times n}$ there exist an invertible lower triangular M and a signed matching matrix P such that*

$$A = M\,P\,M^T.$$

The decomposition can be computed in $O(n^3)$ arithmetic operations.

The proofs are standard elimination arguments closely related to the Bruhat and generalized Bruhat decompositions; for completeness, and to fix the exact normalization used here, they are given in Section Appendix B.

5.1. Canonicity

The matching produced by the reductions is not an artifact of the construction: it is determined by the rank profile of A. For $1 \le i, j \le n$ let

$$r_{ij}(A) := \operatorname{rank}\bigl(A[1{:}i, 1{:}j]\bigr), \qquad \Delta_{ij}(A) := r_{ij} - r_{i-1,j} - r_{i,j-1} + r_{i-1,j-1},$$

with the convention $r_{0j} = r_{i0} = 0$. We call (i, j) a *rank jump* of A if $\Delta_{ij}(A) = 1$.
Proposition 5.5 (canonicity).

1. *If $A = M\Pi N$ as in Lemma 5.3, then*

$$r_{ij}(A) = \#\{(i', j') \in \mathcal{M}(\Pi) {:} i' \le i,\ j' \le j\} \quad \textit{for all } i, j,$$

and consequently $\mathcal{M}(\Pi)$ is exactly the set of rank jumps of A.

2. *If $A = A^T = MPM^T$ as in Lemma 5.4, then the singletons of P are the diagonal rank jumps $\{t{:}\Delta_{tt}(A) = 1\}$ and the pairs of P are the unordered off-diagonal rank jumps $\{\{i, j\}{:}i \ne j,\ \Delta_{ij}(A) = 1\}$.*

Thus, for symmetric A, the unsymmetric normal form contains the two matched pairs (i, j) and (j, i) corresponding to each symmetric pair (i, j) of P, while each singleton of P gives one diagonal matched pair. This is the rank-profile dictionary behind the equivalence between the LU and LDL^T criteria.

6. The criteria on the normal forms are prefix conditions

On a matching matrix, all three nullities entering the criteria are counts of simple combinatorial objects, and the criteria collapse to conditions on prefixes of the index set.

6.1. The unsymmetric normal form

Let Π be a matching matrix with matching $\mathcal{M}$. Relative to the cut after index k, classify the matched pairs and define

$$\begin{aligned}
\iota_k &:= \#\{(i,j) \in \mathcal{M} : i \le k \text{ and } j \le k\} && \text{(pairs inside the leading block)},\\
\chi_k^R &:= \#\{(i,j) \in \mathcal{M} : i \le k < j\} && \text{(pairs crossing the cut row-first)},\\
\chi_k^C &:= \#\{(i,j) \in \mathcal{M} : j \le k < i\} && \text{(pairs crossing the cut column-first)},\\
r_k^0 &:= \#\{\text{zero rows} \le k\}, \qquad c_k^0 := \#\{\text{zero columns} \le k\}.
\end{aligned}$$

Lemma 6.1 (nullities of a matching matrix). *For every k,*

$$\begin{aligned}
\operatorname{null}(\Pi[1{:}k, 1{:}k]) &= r_k^0 + \chi_k^R = c_k^0 + \chi_k^C, \qquad (6.1)\\
\operatorname{null}(\Pi[:, 1{:}k]) &= c_k^0,\\
\operatorname{null}(\Pi[1{:}k, :]^T) &= r_k^0.
\end{aligned}$$

In particular the bookkeeping identity $r_k^0 + \chi_k^R = c_k^0 + \chi_k^C$ *holds.*

Proof. The nonzero columns of the $n \times k$ block $\Pi[:, 1{:}k]$ are the vectors e_i over the matched pairs with $j \le k$; they occupy distinct rows and are therefore linearly independent, while the remaining c_k^0 columns vanish. Hence $\operatorname{rank}(\Pi[:, 1{:}k]) = k - c_k^0$, giving the second formula, and the third follows by applying the same argument to Π^T, whose matching is the transpose of $\mathcal{M}$.

The leading block $\Pi[1{:}k, 1{:}k]$ is itself a matching matrix whose matched pairs are the ι_k pairs inside the block, so $\operatorname{rank}(\Pi[1{:}k, 1{:}k]) = \iota_k$ (as in Proposition 5.5, its nonzero columns are independent). Classify the k row indices $\le k$: each is a zero row, or belongs to a pair with $j \le k$, or to a pair with $j > k$; the classes are disjoint and exhaust, so $k = r_k^0 + \iota_k + \chi_k^R$ and $\operatorname{null}(\Pi[1{:}k, 1{:}k]) = k - \iota_k = r_k^0 + \chi_k^R$. Classifying the column indices instead gives $k = c_k^0 + \iota_k + \chi_k^C$ and the alternative expression, whence the bookkeeping identity. □

Lemma 6.2 (prefix form of the LU criterion). *For a matching matrix Π, criterion (4.1) at index k is equivalent to each of*

(a) $\chi_k^R \le c_k^0$, *(b)* $\chi_k^C \le r_k^0$, *(c)* $\nu_k := \#\{(i,j) \in \mathcal{M} : \min(i,j) \le k\} \le k$.

Proof. Substituting (6.1) into (4.1): using the first expression for the leading nullity, $r_k^0 + \chi_k^R \le c_k^0 + r_k^0$, i.e. (a); using the second, $c_k^0 + \chi_k^C \le c_k^0 + r_k^0$, i.e. (b). By the bookkeeping identity, (a) and (b) are equivalent (they have equal slack). For (c): a pair has $\min(i,j) \le k$ iff it is counted by ι_k, χ_k^R, or χ_k^C, so $\nu_k = \iota_k + \chi_k^R + \chi_k^C$; substituting $\iota_k = k - r_k^0 - \chi_k^R$ from the proof of Lemma 6.1 gives $\nu_k \le k \iff \chi_k^C \le r_k^0$, which is (b). ☐

6.2. *The symmetric normal form*

Let P be a signed matching matrix. Define

$$\zeta_k{:} = \#\{\text{isolated zero indices} \le k\}, \qquad \chi_k{:} = \#\{\text{pairs } (i,j) \text{ of } P{:} i \le k < j\}.$$

Lemma 6.3 (nullities of a signed matching matrix). *For every k,*

$$\mathrm{null}(P[1{:}k, 1{:}k]) = \zeta_k + \chi_k, \qquad \mathrm{null}(P[:, 1{:}k]) = \zeta_k.$$

Proof. The columns $t \le k$ of the $n \times k$ block $P[:, 1{:}k]$ are: zero for isolated t; $\gamma_t e_t$ for singletons; for a pair (i,j) of P, column i of P equals e_j and column j equals e_i (whichever of the two has index $\le k$ is included). Distinct blocks occupy disjoint row sets, and within a pair with both $i,j \le k$ the two columns e_j and e_i are independent (restricted to rows $\{i,j\}$ they form the pair block, of determinant -1). Hence the nonzero columns of $P[:, 1{:}k]$ are independent, exactly the isolated columns vanish, and $\mathrm{null}(P[:, 1{:}k]) = \zeta_k$.

The leading block $P[1{:}k, 1{:}k]$ is block diagonal (up to symmetric permutation) with blocks: zeros at isolated indices (ζ_k of them); (γ_t) at singletons (rank 1); full pair blocks when $j \le k$ (rank 2); and, for each crossing pair $i \le k < j$, the single row/column i whose entries within the block all vanish (its only nonzero sits in column $j > k$) — contributing one null direction each. Hence $\mathrm{null}(P[1{:}k, 1{:}k]) = \zeta_k + \chi_k$. ☐

Lemma 6.4 (prefix form of the LDL^T criterion). *For a signed matching matrix P, criterion (4.2) at index k is equivalent to*

$$\chi_k \ \le\ \zeta_k. \tag{6.2}$$

Proof. Substitute Lemma 6.3 into (4.2): $\zeta_k + \chi_k \le 2\zeta_k$. ☐

Remark 6.5. *Under the rank-profile dictionary of Proposition 5.5 — a pair (i,j) of P corresponds to the two matched pairs (i,j) and (j,i) of the unsymmetric normal form* Π *of the same symmetric A, an isolated zero to a common zero row and column — we have $\chi_k^R = \chi_k^C = \chi_k$ and $r_k^0 = c_k^0 = \zeta_k$, and conditions (a), (b) of Lemma 6.2 both become (6.2). The combinatorial obstruction is thus literally the same in the two problems.*

7. Explicit factorization of the normal forms

We now factor the normal forms under the prefix conditions of Section 6. Both constructions build the factorization as a sum of rank-one terms indexed by *slots* $t \in \{1, \dots, n\}$: for $A = LU$,

$$LU = \sum_{t=1}^{n} \ell_t \, u_t^T, \tag{7.1}$$

where ℓ_t is column t of L (supported on rows $\geq t$) and u_t^T is row t of U (supported on columns $\geq t$); for $A = LDL^T$,

$$LDL^T = \sum_{t=1}^{n} d_t \, \ell_t \ell_t^T. \tag{7.2}$$

The difference between the two problems is now plain: in (7.1) each term is an arbitrary rank-one matrix with the required support, whereas in (7.2) each term is *symmetric* rank-one. An off-diagonal exchange block $e_i e_j^T + e_j e_i^T$ has rank 2, so no single symmetric rank-one term realizes it: it must be written as a cancelling combination of at least two such terms. Each pair of the normal form therefore consumes *two* slots in the symmetric problem, and the prefix condition supplies exactly enough of them.

7.1. LU of a matching matrix: greedy slot assignment

THEOREM 7.1. *Let* Π *be a matching matrix satisfying the prefix condition* $\nu_k \leq k$ *for all* k *(Lemma 6.2(c)). Then there is an injection*

$$\sigma\colon \mathcal{M}(\Pi) \hookrightarrow \{1, \dots, n\}, \qquad \sigma(i,j) \leq \min(i,j) \quad \textit{for every } (i,j) \in \mathcal{M}(\Pi), \tag{7.3}$$

computable by a greedy pass, and setting, for each $(i,j) \in \mathcal{M}$ *with* $t = \sigma(i,j)$,

$$\ell_t \colon = e_i, \qquad u_t \colon = e_j,$$

and $\ell_t = u_t = 0$ *for slots* $t \notin \operatorname{range}(\sigma)$, *the matrices* $L = (\ell_1, \dots, \ell_n)$ *and* U *with rows* $u_1^T, \dots, u_n^T$ *are lower and upper triangular, respectively, and* $LU = \Pi$.

Proof. The assignment suffices. Given σ as in (7.3), triangularity is immediate: $\ell_t = e_i$ has its support at row $i \geq \min(i,j) \geq t$, and u_t at column $j \geq t$. Since σ is injective, each matched pair contributes its own term in (7.1) and empty slots contribute zero, so $LU = \sum_{(i,j)\in\mathcal{M}} e_i e_j^T = \Pi$. Note that both diagonals may carry zeros: slot t has $L[t,t] = 1$ only if $i = t$, and $U[t,t] \neq 0$ only if $j = t$.

The assignment exists. Process the pairs in order of nondecreasing $m(i,j) := \min(i,j)$, ties broken arbitrarily, maintaining the set of free slots (initially $\{1,\dots,n\}$); assign to each pair the largest free slot $t \le m(i,j)$ (the argument below is valid for *any* choice of free slot $\le m(i,j)$; we fix the largest for definiteness). Suppose the greedy fails at a pair with bound $m := m(i,j)$: all slots $1,\dots,m$ are occupied. Every occupied slot $s \le m$ was assigned to a previously processed pair, whose bound was $\le m$ (bounds are nondecreasing along the pass). Together with the current pair, this exhibits $m+1$ pairs with min $\le m$, i.e. $\nu_m \ge m+1$, contradicting the prefix condition. Hence the greedy assigns every pair and produces σ. $\square$

REMARK 7.2 (WHICH PAIRS NEED REROUTING). *Order the pairs by their bound $m(i,j)$. A diagonal pair (t,t) can take its own slot t; a pair (i,j) with $i<j$ (an "above" pair) would naturally take slot i, and a "below" pair (i,j) with $i>j$ slot j. Conflicts occur exactly at* crossing indices *t that are simultaneously the row of an above pair (t,j) and the column of a below pair (i,t): one of the two must be rerouted to an earlier free slot, whose availability is exactly what the prefix condition guarantees — in the leanest case the free slot comes from an index that is both a zero row and a zero column. The exchange matrix A_2 of (1.1) is the minimal example: index 2 is crossing, and the pair $(2,3)$ (or $(3,2)$) is rerouted through slot 1, whose row and column are zero. This is the precise sense in which zero diagonal entries are a* resource *for unpivoted factorization.*

7.2. LDL^T of a signed matching matrix: pair splitting

Throughout this subsection P is a signed matching matrix satisfying $\chi_k \le \zeta_k$ for all k (Lemma 6.4). Thus every pair block has the normalized real form $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$.

The symmetric slot assignment rests on a single identity: a pair block is a difference of two symmetric rank-one terms sharing the same leading row,

$$e_i e_j^T + e_j e_i^T = uu^T - vv^T, \qquad u = e_i + \tfrac{1}{2}e_j, \quad v = e_i - \tfrac{1}{2}e_j, \tag{7.4}$$

both terms supported on $\{i,j\}$.

THEOREM 7.3. *Let P be a signed matching matrix with $\chi_k \le \zeta_k$ for all k. Then there are a lower triangular S and a diagonal E with*

$$P = SES^T.$$

Proof. Form a list of *items*: each singleton t is one item with *bound* t, carrying the column e_t and the weight γ_t; each pair (i,j) is two items with bound i, carrying the columns u,v of (7.4) and the weights $+1$, -1. Suppose the items are placed in pairwise distinct slots, each item in a slot $\leq$ its bound; let ℓ_t and d_t be the column and weight of the item in slot t (with $\ell_t = 0$, $d_t = 0$ for unused slots), and set $S = (\ell_1, \ldots, \ell_n)$, $E = \operatorname{diag}(d_1, \ldots, d_n)$. Every item's column is supported on rows $\geq$ its bound $\geq$ its slot, so S is lower triangular, and

$$SES^T = \sum_{\text{singletons } t} \gamma_t \, e_t e_t^T + \sum_{\text{pairs } (i,j)} \left(uu^T - vv^T\right) = P,$$

since each block of P is reproduced exactly once by (7.4).

Such a placement exists. With s_k the number of singletons $\leq k$ and ι_k the number of pairs (i, j) with $j \leq k$, classifying the indices $\leq k$ into isolated zeros, singletons, pair indices with $j \leq k$, and pair indices with $i \leq k < j$ gives $k = \zeta_k + s_k + 2\iota_k + \chi_k$, so

$$\#\{\text{items with bound} \leq k\} = s_k + 2(\iota_k + \chi_k) = k - (\zeta_k - \chi_k) \ \leq \ k \iff \chi_k \leq \zeta_k.$$

Order the items by nondecreasing bound and place the tth item in slot t: if some item's bound were smaller than its slot t, the items with bound $\leq t - 1$ would number at least t, contradicting the count just displayed at $k = t - 1$. □

Remark 7.4 (threads: a division-free alternative). *The split (7.4) divides by* 2. *An alternative construction avoids all division and realizes each pair by a single column, at the price of one extra column per group: partition the pairs into* threads $r < i_1 < j_1 < i_2 < j_2 < \cdots < i_m < j_m$, *with r an isolated zero index and (i_q, j_q) pairs of P — a left-to-right scan that attaches each pair to a free thread and roots new threads at isolated zeros succeeds under $\chi_k \leq \zeta_k$ — and, with the partial sums $s_q = -(e_{i_q} + e_{i_{q+1}} + \cdots + e_{i_m})$, take the columns and alternating weights*

$$\ell_r = s_1, \ d_r = 1; \qquad \ell_{i_q} = s_q + e_{j_q}, \ d_{i_q} = -1; \qquad \ell_{j_q} = s_{q+1} + e_{j_q}, \ d_{j_q} = 1.$$

The sum $\sum_t d_t \ell_t \ell_t^T$ telescopes to the thread's pair blocks. This variant survives in characteristic 2 *(where (7.4) is unavailable) and keeps each column inside its thread, but it spends one column on every root, so it is not column-minimal; see Section 8.*

8. Sparsity and rank-revealing structure

Unpivoted factorizations of a singular matrix are far from unique, so no sparsity pattern can be forced on *every* factorization. Nevertheless, the normal-form construction can be steered to a factorization whose sparsity pattern is canonical: its envelope is read off the rank-jump matching of A, which by Proposition 5.5 is an invariant of A, and it is optimal among all factorizations in a precise leading-block sense. This parallels the rank-revealing and sparsity theorems of [7], with two differences: no internal permutations appear, and the envelope is expressed directly through the rank jumps. The dictionary between the two formulations is that the matched rows of Π are exactly the rows of A that are linearly independent of their predecessors, and likewise for columns (specialize Proposition 5.5 to the rank functions $k \mapsto \mathrm{rank}(A[1{:}k,:])$ and $k \mapsto \mathrm{rank}(A[:,1{:}k])$).

8.1. The unsymmetric factors

Let A satisfy (4.1), let $\mathcal{M} = \{(i_1, j_1), \ldots, (i_r, j_r)\}$ be its rank-jump matching, and note $r = \mathrm{rank}(A)$. Order the pairs so that the bounds $m_t\!: = \min(i_t, j_t)$ are nondecreasing, with any fixed tie-breaking rule (at most two jumps share a bound: one of the form (m, j) and one of the form (i, m)). The prefix condition of Lemma 6.2(c) states $\nu_k \le k$ for all k, which is equivalent to $m_t \ge t$ for all t: if $m_t < t$ then $\nu_{m_t} \ge t > m_t$, and if $\nu_k > k$ then $m_{k+1} \le k$. Hence assigning the tth pair to slot t is an admissible assignment in Theorem 7.1; we call the resulting factorization, with all matched scalars absorbed into the upper factor, the *sorted* factorization.

Theorem 8.1 (canonical sparsity of the LU factors). *Let A satisfy (4.1). The sorted factorization $A = LU$ has the following properties.*

1. Rank-revealing: *only the first r columns of L and the first r rows of U are nonzero: $L[:, r+1{:}] = 0$ and $U[r+1{:}, :] = 0$.*
2. Canonical envelope: *for $t \le r$, column t of L vanishes above row i_t and $L[i_t, t] = 1$; row t of U vanishes left of column j_t and $U[t, j_t] \ne 0$.*
3. Diagonals: *$L[t, t] = 1$ if $i_t = t$ and $L[t, t] = 0$ otherwise; $U[t, t] \ne 0$ if and only if $j_t = t$.*
4. Simultaneous envelope optimality: *for* every *unpivoted factorization $A = \widetilde{L}\widetilde{U}$ and every k,*

$$\begin{aligned} \#\{t{:}\widetilde{L}[1{:}k, t] \ne 0\} \;&\ge\; \mathrm{rank}(A[1{:}k, :]) = k - r_k^0, \\ \#\{t{:}\widetilde{U}[t, 1{:}k] \ne 0\} \;&\ge\; \mathrm{rank}(A[:, 1{:}k]) = k - c_k^0, \end{aligned}$$

   and the sorted factorization attains both bounds for every k simultaneously.

The data $(i_t, j_t)_{t \le r}$ — hence the entire envelope pattern — is uniquely determined by A, up to the tie-breaking rule.

Proof. Write $A = M\Pi N$ as in Lemma 5.3. Since each matched scalar occupies its own column of the pre-normalized pattern, all scalars can be pulled into a diagonal factor on the right and absorbed into N; with this convention M is unit lower triangular, N is invertible upper triangular, and $\Pi = \sum_t e_{i_t} e_{j_t}^T$. The sorted assignment gives $L = M\left(\sum_{t\le r} e_{i_t} e_t^T\right)$ and $U = \left(\sum_{t\le r} e_t e_{j_t}^T\right) N$, i.e.,

$$L[:,t] = M[:,i_t], \qquad U[t,:] = N[j_t,:] \qquad (t \le r),$$

and $L[:,t] = 0$, $U[t,:] = 0$ for $t > r$. Claims (1)–(3) follow from the triangularity of M and N: $M[:,i_t]$ vanishes above row i_t with $M[i_t,i_t] = 1$, and $M[t,i_t] = 0$ whenever $i_t > t$; similarly for N, whose diagonal entries are nonzero.

For (4), the lower bounds hold for every factorization: from $A[1{:}k,:] = \widetilde{L}[1{:}k,:]\,\widetilde{U}$,

$$\operatorname{rank}(A[1{:}k,:]) \le \operatorname{rank}(\widetilde{L}[1{:}k,:]) \le \#\{t{:}\widetilde{L}[1{:}k,t] \ne 0\},$$

and dually for $\widetilde{U}$. The sorted factorization attains them: $L[1{:}k,t] \ne 0$ if and only if $i_t \le k$, and the number of matched rows $\le k$ equals $\operatorname{rank}(A[1{:}k,:]) = k - r_k^0$ (as in the proof of Lemma 6.1, transported to A by Lemma Appendix A.4). □

8.2. The symmetric factors: a rank-revealing LDL^T

The construction of Theorem 7.3 becomes rank-revealing when its items are placed by the sorted assignment used in its proof. Recall from that proof the counts s_k (singletons $\le k$) and ι_k (pairs (i,j) with $j \le k$), and the index identity $k = \zeta_k + s_k + 2\iota_k + \chi_k$.

Theorem 8.2 (rank-revealing LDL^T). *Let $A = A^T$ satisfy (4.2) and let $r = \operatorname{rank}(A)$. There is an unpivoted factorization $A = LDL^T$ such that:*

1. *only the first r columns of L are nonzero, and D has exactly r nonzero entries, namely $D[t,t] \in \{\pm 1\}$ for $t \le r$, of which exactly $n_+(A)$ equal $+1$ and $n_-(A)$ equal -1, where $n_\pm(A)$ count the positive and negative eigenvalues of A: the diagonal factor reveals both the rank and the inertia of A;*
2. *each singleton t of the signed matching form of A contributes one column with support starting at row t, and each pair (i,j) contributes two columns with support starting at row i, with D-entries $+1$ and -1;*
3. *for every k,*

$$\#\{t{:}L[1{:}k,t] \ne 0\} = s_k + 2\iota_k + 2\chi_k = k - (\zeta_k - \chi_k). \tag{8.1}$$

Proof. By Lemma 5.4, $A = MPM^T$; let $P = SES^T$ be the factorization constructed in the proof of Theorem 7.3, with the items sorted by nondecreasing bound and the tth item placed in slot t, and set $L = MS$, $D = E$. The items number $s + 2p$ in total (s singletons, p pairs), and $s + 2p = \operatorname{rank}(P) = \operatorname{rank}(A) = r$, since each singleton block has rank 1 and each pair block rank 2. Hence exactly the slots $1,\dots,r$ are used, and D carries the item weights, which lie in $\{\pm 1\}$: $\gamma_t = \pm 1$ for singletons, $+1$ and -1 for the two halves of each pair. Since M is invertible lower triangular, multiplication by M preserves column supports and their leading rows, giving (1) and (2), except for the inertia claim, which follows from Sylvester's law: $A = MPM^T$ with M invertible, so A and P have the same inertia; up to a symmetric permutation, P is the direct sum of its blocks, whose inertias are $(\operatorname{sgn}\gamma_t)$ for singletons and one positive, one negative eigenvalue (± 1) for each pair block; and the weights placed in D — γ_t per singleton, $+1$ and -1 per pair — match these counts exactly. For (8.1): the columns of L meeting rows $1,\dots,k$ are exactly the items with bound $\le k$, which number $s_k + 2(\iota_k + \chi_k) = k - (\zeta_k - \chi_k)$, as counted in the proof of Theorem 7.3. $\square$

The count (8.1) exceeds the naive bound $\operatorname{rank}(A[1{:}k,:]) = s_k + 2\iota_k + \chi_k$ by χ_k, and this excess is unavoidable: crossing pairs genuinely cost two columns each at the cut.

Proposition 8.3 (envelope lower bound for LDL^T). *Let $A = A^T = LDL^T$ be any unpivoted factorization. Then for every k,*

$$\#\{t{:}L[1{:}k,t] \neq 0\} \;\ge\; s_k + 2\iota_k + 2\chi_k = k - (\zeta_k - \chi_k),$$

so the factorization of Theorem 8.2 is simultaneously optimal for every k. In particular ($k = n$) every factorization has at least r nonzero columns with nonzero diagonal weight, and Theorem 8.2 attains r.

Proof. Discarding columns with $D[t,t] = 0$ changes neither A nor decreases the count, so assume all diagonal weights of the retained columns are nonzero. Fix k, let $C = \{t{:}L[1{:}k,t] \neq 0\}$, $c = |C|$, and let $X = L[1{:}k, C]$ and $X' = L[:, C]$. Columns outside C contribute nothing to the first k rows, so

$$A[1{:}k, 1{:}k] = XD_CX^T, \qquad A[1{:}k,:] = XD_CX'^T.$$

Equip $\mathbb{R}^c$ with the nondegenerate symmetric bilinear form $\langle u, v\rangle = u^T D_C v$, and let $Z \subseteq \mathbb{R}^c$ be the row space of X, of dimension $\rho = \operatorname{rank}(X)$. From the second identity, $\rho \ge \operatorname{rank}(A[1{:}k,:]) = s_k + 2\iota_k + \chi_k$. The Gram matrix of the rows of X is $XD_CX^T = A[1{:}k,1{:}k]$, so the form restricted to Z has rank $\operatorname{rank}(A[1{:}k,1{:}k]) = s_k + 2\iota_k$, and the radical $Z \cap Z^\perp$ has dimension $\rho - (s_k + 2\iota_k)$. Since the form on $\mathbb{R}^c$ is nondegenerate, $\dim Z^\perp = c - \rho$, and the radical is contained in $Z^\perp$, whence

$$c - \rho \;\ge\; \rho - (s_k + 2\iota_k), \qquad \text{i.e.} \qquad c \;\ge\; 2\rho - (s_k + 2\iota_k) \;\ge\; s_k + 2\iota_k + 2\chi_k. \qquad \square$$

Remark 8.4. *The thread variant of Remark 7.4 is not column-minimal — it spends one column on the root of each thread, for $r + (\#threads)$ nonzero columns in total — and its diagonal need not reveal the inertia: when L is singular, Sylvester's law of inertia does not constrain D. Example 9.1 illustrates both phenomena: the split factorization, displayed in (1.2), uses two columns and the diagonal* $\mathrm{diag}(1, -1, 0)$, *attaining (8.1) at every k, while the thread variant uses three columns and the diagonal* $\mathrm{diag}(1, -1, 1)$, *whose inertia differs from that of A_2.*

9. Proofs of the main theorems, and a worked example

Proof (Proof of Theorem 4.1). Necessity is Proposition Appendix A.3. For sufficiency, assume A satisfies (4.1) for all k. By Lemma 5.3, $A = M\Pi N$ with M, N invertible triangular and Π a matching matrix. By Corollary Appendix A.5, Π satisfies (4.1), hence by Lemma 6.2 the prefix condition $\nu_k \leq k$ for all k. By Theorem 7.1 there are lower triangular L_Π and upper triangular U_Π with $L_\Pi U_\Pi = \Pi$. Then

$$A = M\,(L_\Pi U_\Pi)\,N = (ML_\Pi)(U_\Pi N),$$

and ML_Π is lower triangular, $U_\Pi N$ upper triangular. □

Proof (Proof of Theorem 4.2). Necessity is Proposition Appendix A.3. For sufficiency, assume the symmetric A satisfies (4.2) for all k. By Lemma 5.4, $A = MPM^T$ with M invertible lower triangular and P a signed matching matrix. By Corollary Appendix A.5, P satisfies (4.2), hence by Lemma 6.4 the prefix condition $\chi_k \leq \zeta_k$ for all k. By Theorem 7.3, $P = SES^T$ with S lower triangular and E diagonal. Then

$$A = MS\,E\,S^T M^T = (MS)\,E\,(MS)^T,$$

and MS is lower triangular. □

Example 9.1 (the exchange matrix, both ways). *The matrix A_2 of (1.1) shows why zero diagonal entries are a resource rather than a defect. As a matching matrix, $\mathcal{M} = \{(2,3),(3,2)\}$; row* 1 *and column* 1 *are zero. The prefix counts are $\nu_1 = 0 \leq 1$ and $\nu_2 = 2 \leq 2$ (both pairs have* $\min = 2$*), so Theorem 7.1 applies: the greedy assignment gives slot* 2 *to* $(2,3)$ *and reroutes* $(3,2)$ *to slot* 1. *This is exactly the LU factorization displayed in (1.2).*

As a signed matching matrix, A_2 itself has isolated zero 1 *and the normalized exchange block* $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ *on indices* $(2,3)$. *The pair splits by (7.4) into two items with bound* 2, *placed in slots* 1 *and* 2; *this produces exactly the LDL^T factorization displayed in (1.2), routing* *the off-diagonal block through the earlier zero index: the second slot exists because index* 1 *is isolated. The thread variant of Remark 7.4, with the single thread* $1 < 2 < 3$*, gives instead*

$$S = \begin{pmatrix} 0 & 0 & 0 \\ -1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad E = \operatorname{diag}(1, -1, 1), \qquad A_2 = SES^T,$$

with three nonzero columns, and with a diagonal whose inertia $(n_+, n_-) = (2, 1)$ *differs from the inertia* $(1, 1)$ *of* A_2*: when the triangular factor is singular, Sylvester's law of inertia does not constrain the diagonal factor.*

10. Conclusion

We gave a unified treatment of the existence of triangular factorizations without pivoting for possibly singular real matrices. The unsymmetric criterion (Theorem 4.1) recovers the theorem of Okunev and Johnson; the symmetric criterion (Theorem 4.2) — existence of $A = LDL^T$ with truly diagonal D and possibly singular L — appears to be new, as does the corollary that for symmetric matrices unpivoted LU and LDL^T existence coincide. The proof method is uniform: reduce to a matching normal form by triangular equivalence or congruence, transfer the nullity criterion to the normal form where it becomes a prefix condition on the matching, and factor the normal form explicitly by a sorted slot assignment — with each symmetric off-diagonal pair split into a difference of two rank-one terms ((7.4)), and with a division-free telescoping-thread alternative recorded in Remark 7.4.

Several directions remain open.

- *Numerical stability.* The construction selects pivots for algebraic, not numerical, reasons; a backward-stable variant over $\mathbb{R}$ (perhaps with size-aware choices among the admissible slot assignments, which are far from unique) is an open problem, as in [7].
- *Multiplicity.* Dopico, Johnson, and Molera [8] parametrized all LU factorizations of a singular matrix with L unit lower triangular. Parametrizing all unpivoted LDL^T factorizations — equivalently, all slot assignments, splittings, and diagonal choices — would quantify the freedom our constructions exploit.
- *Structure of the factors.* Section 8 settles the column-envelope structure: both factorizations can be made rank-revealing with a canonical, simultaneously optimal envelope. What remains open is the finer structure *inside* the envelope — entrywise sparsity and fill bounds analogous to those of [7], and their interaction with the freedom in the slot assignment.

- *Complex and block variants.* A complex version with transpose, or a Hermitian version with conjugate transpose, should have closely related normal forms. Rectangular and block versions of the criteria are also natural.

Appendix A. Standard rank facts, necessity, and transfer

Lemma Appendix A.1 (Sylvester's rank inequality). *Let $B \in \mathbb{R}^{m\times k}$ and $C \in \mathbb{R}^{k\times p}$. Then*

$$\operatorname{rank}(B) + \operatorname{rank}(C) - k \;\le\; \operatorname{rank}(BC) \;\le\; \min\bigl(\operatorname{rank}(B), \operatorname{rank}(C)\bigr).$$

Proof. The upper bound is standard. For the lower bound, restrict the linear map $x \mapsto Bx$ to $\operatorname{range}(C)$: its image is $\operatorname{range}(BC)$ and its kernel is contained in $\ker(B)$, so $\operatorname{rank}(C) = \dim\operatorname{range}(C) \le \operatorname{rank}(BC) + \operatorname{null}(B) = \operatorname{rank}(BC) + k - \operatorname{rank}(B)$. □

Lemma Appendix A.2 (invariance of rank and nullity). *If X is an $m \times p$ matrix and $G \in \mathbb{R}^{m\times m}$, $H \in \mathbb{R}^{p\times p}$ are invertible, then* $\operatorname{rank}(GXH) = \operatorname{rank}(X)$ *and* $\operatorname{null}(GXH) = \operatorname{null}(X)$.

Proof. Left multiplication by G is an isomorphism of the codomain, so it does not change the dimension of the range. Right multiplication by H is an isomorphism of the domain and sends $\ker(XH)$ bijectively onto $\ker(X)$ via $v \mapsto Hv$; hence $\operatorname{null}(XH) = \operatorname{null}(X)$ and therefore $\operatorname{rank}(XH) = p - \operatorname{null}(XH) = \operatorname{rank}(X)$. Combining the two observations gives the rank identity for GXH, and the nullity identity follows from $\operatorname{null}(\cdot) = p - \operatorname{rank}(\cdot)$. □

Proposition Appendix A.3 (necessity). *If $A = LU$ with L lower and U upper triangular, then (4.1) holds for every k. Consequently, if $A = A^T$ and $A = LDL^T$ with L lower triangular and D diagonal, then (4.2) holds for every k.*

Proof. Fix k and partition

$$L = \begin{pmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{pmatrix}, \qquad U = \begin{pmatrix} U_{11} & U_{12} \\ 0 & U_{22} \end{pmatrix},$$

with L_{11}, U_{11} of size $k \times k$. Reading off the leading block, the first k columns, and the first k rows of $A = LU$, and using (3.1),

$$A[1{:}k, 1{:}k] = L_{11}U_{11}, \qquad A[:, 1{:}k] = \begin{pmatrix} L_{11} \\ L_{21} \end{pmatrix} U_{11}, \qquad A[1{:}k, :] = L_{11}\begin{pmatrix} U_{11} & U_{12} \end{pmatrix}.$$

The second identity gives $\mathrm{rank}(A[:,1{:}k]) \le \mathrm{rank}(U_{11})$ and the third gives $\mathrm{rank}(A[1{:}k,:]) \le \mathrm{rank}(L_{11})$. Applying Lemma Appendix A.1 to $A[1{:}k,1{:}k] = L_{11}U_{11}$,

$$\mathrm{rank}(A[1{:}k,1{:}k]) \ \ge\ \mathrm{rank}(L_{11}) + \mathrm{rank}(U_{11}) - k \ \ge\ \mathrm{rank}(A[1{:}k,:]) + \mathrm{rank}(A[:,1{:}k]) - k.$$

Subtracting each side from k and using $\mathrm{null}(X) = k - \mathrm{rank}(X)$ for the three k-column blocks $A[1{:}k,1{:}k]$, $A[:,1{:}k]$, $A[1{:}k,:]^T$ yields

$$\mathrm{null}(A[1{:}k,1{:}k]) \ \le\ 2k - \mathrm{rank}(A[1{:}k,:]) - \mathrm{rank}(A[:,1{:}k]) \ =\ \mathrm{null}(A[1{:}k,:]^T) + \mathrm{null}(A[:,1{:}k]),$$

which is (4.1).

For the symmetric statement, observe that $A = LDL^T = L\cdot(DL^T)$ is an LU factorization (DL^T is upper triangular), so (4.1) holds; since $A = A^T$ gives $A[1{:}k,:]^T = A[:,1{:}k]$, the right-hand side of (4.1) equals $2\,\mathrm{null}(A[:,1{:}k])$, which is (4.2). □

Lemma Appendix A.4 (transfer). *Let $M \in \mathbb{R}^{n\times n}$ be invertible lower triangular, $N \in \mathbb{R}^{n\times n}$ invertible upper triangular, and $B = MAN$. Then for every $k = 1,\dots,n$,*

$$\mathrm{null}(B[1{:}k,1{:}k]) = \mathrm{null}(A[1{:}k,1{:}k]), \tag{A.1}$$
$$\mathrm{null}(B[:,1{:}k]) = \mathrm{null}(A[:,1{:}k]), \tag{A.2}$$
$$\mathrm{null}(B[1{:}k,:]^T) = \mathrm{null}(A[1{:}k,:]^T). \tag{A.3}$$

Consequently A satisfies (4.1) for every k if and only if B does; and, taking $N = M^T$, a symmetric A satisfies (4.2) for every k if and only if the congruent matrix $B = MAM^T$ does.

Proof. Write $M_k = M[1{:}k,1{:}k]$ and $N_k = N[1{:}k,1{:}k]$; both are invertible (Section 3). Using (3.1) three times:

- $B[1{:}k,1{:}k] \ =\ M[1{:}k,:]\,A\,N[:,1{:}k] \ =\ M_k\,A[1{:}k,1{:}k]\,N_k$, since the zero blocks in (3.1) suppress the rows of A beyond k on the left and the columns beyond k on the right;
- $B[:,1{:}k] = MA\,N[:,1{:}k] = M\,A[:,1{:}k]\,N_k$;
- $B[1{:}k,:] = M[1{:}k,:]\,AN = M_k\,A[1{:}k,:]\,N$, and hence $B[1{:}k,:]^T = N^T\,A[1{:}k,:]^T\,M_k^T$.

In each identity the outer factors are invertible and the two sides have the same number of columns, so Lemma Appendix A.2 gives (A.1) to (A.3). The congruence case follows because M^T is invertible upper triangular, and for symmetric A the quantity $\mathrm{null}(A[1{:}k,:]^T)$ in (4.1) equals $\mathrm{null}(A[:,1{:}k])$. □

Combining Lemma Appendix A.4 with the reductions of Section 5 (applied with M^{-1} and N^{-1}, which are again invertible triangular):

Corollary Appendix A.5. *Let $A = M\Pi N$ as in Lemma 5.3. Then A satisfies (4.1) for every k if and only if Π does. Let $A = MPM^T$ as in Lemma 5.4. Then A satisfies (4.2) for every k if and only if P does.*

Proof. Apply Lemma Appendix A.4 to $\Pi = M^{-1}AN^{-1}$ in the LU case and to $P = M^{-1}AM^{-T}$ in the symmetric case. The inverses of invertible lower and upper triangular matrices are triangular of the same kind, so the hypotheses of the transfer lemma are satisfied. $\square$

Appendix B. Proofs for the matching normal forms

Proof (Proof of Lemma 5.3). We induct on the leading order k. For $k = 1$, take $\Pi = (0)$ if $a_{11} = 0$, and take $\Pi = (1)$ with the scalar a_{11} absorbed into one of the two outer factors if $a_{11} \neq 0$. Assume $A[1{:}k-1, 1{:}k-1] = M\Pi N$, with M invertible lower triangular, N invertible upper triangular, and Π a matching matrix. Border the next leading block as

$$A[1{:}k, 1{:}k] = \begin{pmatrix} A[1{:}k-1, 1{:}k-1] & a \\ c^T & b \end{pmatrix}.$$

With $\widehat{a} = M^{-1}a$ and $\widehat{c}^T = c^T N^{-1}$, the middle bordered matrix is

$$X = \begin{pmatrix} \Pi & \widehat{a} \\ \widehat{c}^T & b \end{pmatrix}.$$

First remove from the border the components already explained by matched rows and columns. For every matched pair (i, j) set $x_j = \widehat{a}_i$ and $y_i = \widehat{c}_j$, with all other components zero. Then

$$e{:} = \widehat{a} - \Pi x, \qquad g^T{:} = \widehat{c}^T - y^T\Pi$$

are supported on zero rows and zero columns of Π, respectively, and $y^T e = g^T x = 0$. Hence, with $f = b - y^T\Pi x$,

$$X = \begin{pmatrix} I & 0 \\ y^T & 1 \end{pmatrix} \begin{pmatrix} \Pi & e \\ g^T & f \end{pmatrix} \begin{pmatrix} I & x \\ 0 & 1 \end{pmatrix}.$$

These two outer factors are unit lower and unit upper triangular.

It remains to turn the residual border into new matched pairs. If $e \neq 0$, let $i^\star$ be the first nonzero index of e and let $\alpha = e_{i^\star}$. Since e is supported on zero rows, the unit lower triangular matrix

$$R = I + (\alpha^{-1}e - u_{i^\star})u_{i^\star}^T$$

satisfies $R\Pi = \Pi$ and $Ru_{i^\star} = \alpha^{-1}e$, converting the residual column to $\alpha u_{i^\star}$. If $g \neq 0$, an analogous unit upper triangular matrix converts the residual row to $\gamma u_{j^\star}^T$, where $j^\star$ is a zero column of Π and $\gamma = g_{j^\star} \neq 0$.

After these operations, the only possible nonzeros outside Π are the entries $(i^\star, k)$, $(k, j^\star)$, and the corner (k, k). If a residual column is present, the row $i^\star$ has only the entry $(i^\star, k) = \alpha$, so a final unit lower triangular operation subtracts the corner through that row. If no residual column is present but a residual row is present, the analogous unit upper triangular operation removes the corner through column $j^\star$. If neither residual is present, the corner itself is either a new diagonal matched pair (k, k) or zero.

The support pattern that remains is a matching pattern: newly used rows and columns were zero in the old matching, and the new index k is used at most once as a row and at most once as a column. Replace every new nonzero in this pattern by 1 to form the new matching matrix $\Pi^{(k)}$, and absorb the corresponding nonzero scalars into diagonal factors on the left and right. Concretely, the scalar in column k is absorbed into the new diagonal entry of the upper triangular factor, the scalar in row k into the new diagonal entry of the lower triangular factor, and, in the diagonal-pair case, the corner scalar into either side. These diagonal scalings are invertible whenever a new pair is present. This completes the induction. The construction uses $O(k^2)$ operations at the kth step, hence $O(n^3)$ operations overall. □

Proof (Proof of Lemma 5.4). Again we induct on k. For $k = 1$, take $P = (0)$ if $a_{11} = 0$; otherwise scale by $|a_{11}|^{-1/2}$, i.e. take $M = (|a_{11}|^{1/2})$ and $P = (\operatorname{sgn} a_{11})$. Assume $A[1{:}k-1, 1{:}k-1] = MPM^T$, with M lower triangular and P a symmetric matching matrix in the normalized real sense of Definition 5.2. For the next border write

$$A[1{:}k, 1{:}k] = \begin{pmatrix} A[1{:}k-1, 1{:}k-1] & a \\ a^T & b \end{pmatrix}, \qquad c = M^{-1}a.$$

Thus the middle bordered matrix is

$$X = \begin{pmatrix} P & c \\ c^T & b \end{pmatrix}.$$

Since every nonzero block of P is nonsingular, choose ξ supported on the singleton and pair indices so that $P\xi$ agrees with c on those indices. Explicitly, on a singleton t take $\xi_t = \gamma_t c_t$ (so $\gamma_t \xi_t = c_t$, using $\gamma_t^2 = 1$); on a normalized pair (i, j) take $\xi_j = c_i$ and $\xi_i = c_j$. Then $e{:} = c - P\xi$ is supported on isolated zero indices, so $\xi^T e = 0$, and with $f = b - \xi^T P\xi$ we have the congruence

$$X = \begin{pmatrix} I & 0 \\ \xi^T & 1 \end{pmatrix} \begin{pmatrix} P & e \\ e^T & f \end{pmatrix} \begin{pmatrix} I & \xi \\ 0 & 1 \end{pmatrix}.$$

If $e = 0$, the middle factor is $P \oplus (f)$. If $f = 0$, the new index is an isolated zero. If $f \neq 0$, scale the new index by $|f|^{-1/2}$ — a diagonal, hence lower triangular, congruence — so that the appended singleton carries $\gamma_k = \operatorname{sgn}(f) = \pm 1$. If $e \neq 0$, let $i^\star$ be the first nonzero index of e and let $\alpha = e_{i^\star}$. The index $i^\star$ is an isolated zero of P. The lower triangular matrix

$$R = I + (\alpha^{-1}e - u_{i^\star})u_{i^\star}^T$$

satisfies $RPR^T = P$ and $Ru_{i^\star} = \alpha^{-1}e$, so the residual block is congruent to one with pair block $\left(\begin{smallmatrix} 0 & \alpha \\ \alpha & f \end{smallmatrix}\right)$ on indices $(i^\star, k)$. Normalize this real block as follows. Scale the new index k by α^{-1}, making the off-diagonal entry equal to 1 and the lower-right entry f/α^2. Then replace row and column k by row and column k plus

$$\tau = -\frac{f}{2\alpha^2}$$

times row and column $i^\star$. This lower-triangular shear leaves the $(i^\star, i^\star)$ entry equal to zero and the off-diagonal entry equal to 1, while changing the lower-right entry to 0. Hence the new block is exactly $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and the induction closes. $\square$

Proof (Proof of Proposition 5.5). For the unsymmetric normal form, (3.1) gives

$$A[1{:}i, 1{:}j] = M_i \, \Pi[1{:}i, 1{:}j] \, N_j$$

with M_i and N_j invertible. Therefore $r_{ij}(A) = r_{ij}(\Pi)$ by Lemma Appendix A.2. The nonzero columns of $\Pi[1{:}i, 1{:}j]$ are the vectors $e_{i'}$ for the matched pairs with $i' \leq i$ and $j' \leq j$; they occupy distinct rows and are linearly independent. This proves the displayed rank formula in part (1). Taking the mixed differences of that formula shows that the rank jumps are exactly the matched pairs.

The symmetric statement follows from the same rank-invariance argument with $N = M^T$. A singleton t contributes the rank function $\mathbf{1}_{\{i \geq t,\ j \geq t\}}$, hence one jump at (t, t). A normalized pair (i_0, j_0) contributes rank one when exactly one of its off-diagonal entries is present and rank two when the whole 2×2 block is present. Its rank contribution is therefore

$$\mathbf{1}_{\{i \geq i_0,\ j \geq j_0\}} + \mathbf{1}_{\{i \geq j_0,\ j \geq i_0\}},$$

whose mixed differences are the two off-diagonal jumps (i_0, j_0) and (j_0, i_0). Since distinct blocks use disjoint row and column sets, the contributions add, proving part (2). $\square$

Appendix C. Additional example

Example Appendix C.1 (a genuine crossing). *Let* $n = 4$ *and*

$$\Pi = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \mathcal{M} = \{(2,4),(4,2)\},$$

with rows and columns $1, 3$ *zero. Both pairs have* $\min = 2$*, so* $\nu_2 = 2 \le 2$*. The greedy assignment gives slot* 2 *to* $(2,4)$ *and slot* 1 *to* $(4,2)$*:*

$$L = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad U = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad LU = \Pi.$$

Symmetrically, the same crossing pattern is routed through an earlier isolated zero index — the pair needs two slots ≤ 2*, and slots* $1, 2$ *are available — exactly as in Example 9.1.*

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